Arch. Mikrobiol. 67, 16–20 (1969)

Studies on Polyol Metabolism in *Aspergillus niger**

III. Purification and Properties of Sorbitol Dehydrogenase

B. M. Desai, V. V. Modi, and V. K. Shah

Department of Microbiology, Faculty of Science, M. S. University of Baroda, Baroda, India

Received February 28, 1969

Summary. Sorbitol dehydrogenase has been purified about 26 fold from a strain of *Aspergillus niger*, growing on sorbitol as the sole source of carbon. An absolute specificity of this enzyme for sorbitol, fructose, NAD and NADH was observed. The K_m for sorbitol and fructose were found to be 9.8×10^{-5} M and 6.6×10^{-4} M respectively. The enzyme was inhibited by pCMB, NaF and other metal ions studied. The enzyme was slightly activated by Fe^{+++}.

The induction of sorbitol dehydrogenase and changes in its activity during cultivation of the adapted strain of *Aspergillus niger* have been reported in the previous communications (Desai *et al.*, 1967, 1969b). Sorbitol dehydrogenase has been purified from rat liver (Blackley, 1951; McCorkindale and Edson, 1954), *Bacillus subtilis* (Horwitz and Kaplan, 1964) and *Acetobacter suboxydans* (Cummins *et al.*, 1957). This enzyme has not been purified from any mold so far.

The present communication reports partial purification and properties of sorbitol dehydrogenase from *A. niger*.

Materials and Methods

The adapted strain of *A. niger* (Desai *et al.*, 1969a) was grown and the mats were collected on the third day of growth. The growth conditions, preparation of cell free extract and the sorbitol dehydrogenase assay method remained same as described earlier (Desai *et al.*, 1967).

The forward reactions with polyols were carried out by the method of Cummins *et al.* (1957), but $MgCl_2$ was omitted from the system.

The protein was determined by the method described by Lowry *et al.* (1951).

Fractionation of the Cell Free Extract

All the subsequent operations were carried out between 0–4° C. To 90 ml of the high speed supernatent 21.87 g of dry Ammonium sulfate was added slowly with constant stirring and the mixture was held at 0–4° C for 2–4 hours. The

* Part of this work was presented at the All India Conference of Microbiologists held at Baroda, 1968/69.

precipitate was centrifuged down and suspended in 60 ml 0.2 M Tris-HCl buffer pH 8.0. To the suspension 15 ml of $CaPO_4$ gel (10 mg/ml dry wt) was added and held for 30 min. The gel was centrifuged down and the supernatent contained most of the activity. This supernatent was used for all subsequent experiments.

Results and Discussion

The purification of sorbitol dehydrogenase (Table 1) resulted in nearly 26 fold increase in specific activity with 70% recovery. Studies on proportionality of enzyme activity to protein concentration, changes in pH, substrate concentration and NAD concentration are shown in

Table 1. *Purification of sorbitol dehydrogenase*

Fraction extract	Total activity (units)	Total protein (mg)	Specific activity	Recovery (%)
Cell-free	288	468	0.615	100
0—40 $(NH_4)_2SO_4$	216	72	3.0	75
Ca-PO_4 gel supernatent	202	12.75	15.85	70.1

Table 2. *Substrate and coenzyme specificities of sorbitol dehydrogenase*

Polyol/+ NaD/NADP	Specific activity	Sugar/+ NADH/NADPH	Specific activity
Sorbitol + NAD	3.53	Fructose + NADH	91.76
Sorbitol + NADP	nil	Fructose + NADPH	0.00
Ribitol + NAD	nil	Sorbose + NADH	0.00
Ribitol + NADP	nil	Sorbose + NADPH	0.00
Mannitol + NAD	nil	Glucose + NADH/NADPH	0.00
Mannitol + NADP	nil	Mannose + NADH/NADPH	1.05
Arabitol + NAD	nil	Rhamnose + NADH/NAPDH	0.00
Arabitol + NADP	nil	Raffinose + NADH/NADPH	0.00
Xylitol + NAD	nil	Ribose + NADH/NADPH	0.00
Xylitol + NADP	nil	Xylose + NADH/NADPH	0.47
meso-Inositol + NAD	nil	Arabinose + NADH/NADPH	0.00
meso-Inositol + NADP	nil	Trehalose + NADH/NADPH	0.00

Figs. 1, 2, 3 and 4 respectively. The enzyme showed different pH optima for the forward and the backward reaction, pH 10.0 being for the former and pH 9.0 for the latter. This observation is in line with the number of observations for mannitol dehydrogenase from *Lactobacillus brevis* by MARTINEZ *et al.* (1963) and mannitol-1-phosphate dehydrogenase by WOLFF and KAPLAN (1956) and KLUNGSØYR (1966) from *E. coli*.

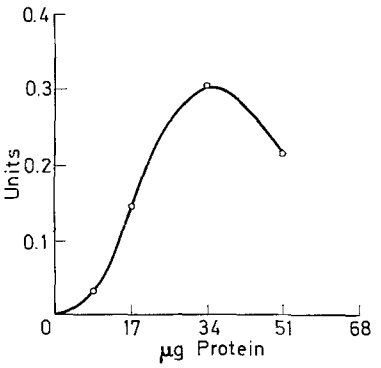


Fig. 1

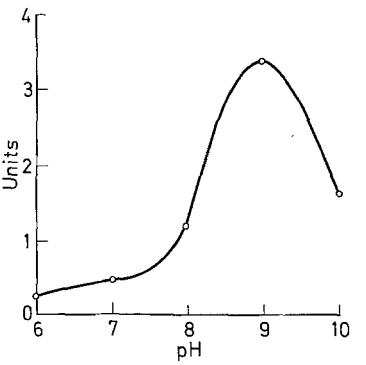


Fig. 2

Fig. 1. Effect of protein concentration on enzyme activity

Fig. 2. Effect of pH on enzyme activity

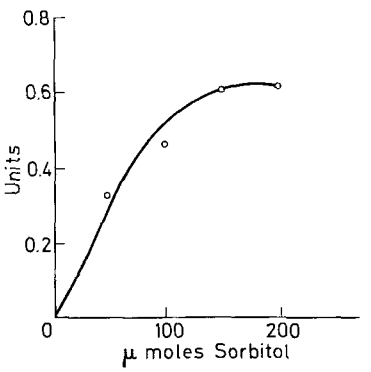


Fig. 3

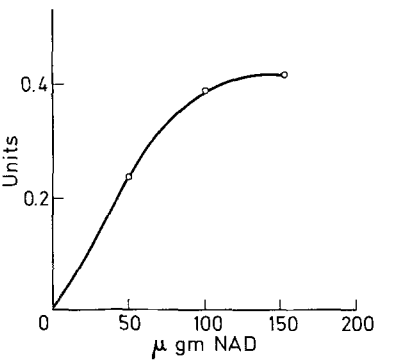


Fig. 4

Fig. 3. Effect of substrate concentration on enzyme activity

Fig. 4. Effect of NAD concentration on enzyme activity

A survey of substrate and coenzyme specificity revealed an absolute specificity of the enzyme for sorbitol, fructose, NAD and NADH (see Table 2). Hence the enzyme catalyzes the only reaction: Sorbitol + NAD^+ $\rightleftharpoons$ Fructose + NADH + H^+ unlike many other non-specific polyol dehydrogenases listed by (TOUSTER and SHAW, 1962; and LEWIS and SMITH, 1967). Slight activities were detected for the backward reactions using mannose and xylose, while no activity was detected for the forward reaction using mannitol and xylitol. Even though mannose and xylose used were of maximum purity available, trace contamination of fructose cannot be overlooked.

Table 3. *Effect of metal ions and various compounds on sorbitol dehydrogenase*

Metal or Inhibitor	Concentration (M)	Activity (%)
Nil	—	100
$CuSO_4$	1.0×10^{-4}	0.00
$HgCl_2$	6.1×10^{-3}	0.00
$ZnSO_4$	2.4×10^{-2}	22.7
$MnSO_4$	9.0×10^{-4}	42.00
$MgCl_2$	6.6×10^{-3}	91.5
$FeCl_3$	2.4×10^{-3}	115.0
GSH	5×10^{-4}	103.0
pCMB	0.5×10^{-4}	0.00
NaF	1.3×10^{-2}	35.2

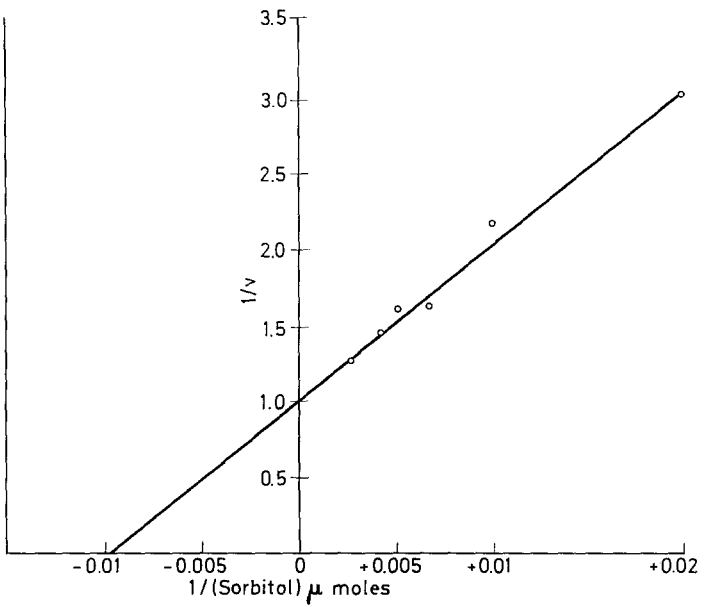


Fig. 5. Double reciprocal plot for sorbitol

The partially purified enzyme was found to be completely inhibited by pCMB, $CuSO_4$ and $HgCl_2$ at the concentrations studied, while NaF, $ZnSO_4$ $MnCl_2$ and $MgCl_2$ inhibited to a lesser extent (see Table 3). Sorbitol dehydrogenase purified by CUMMINS *et al.* (1957) was stimulated by Mg^{++} and Mn^{++} while Zn^{++} and Ca^{++} were ineffective. However, the enzyme from *A. niger* was inhibited by Zn^{++}, Mn^{++}, and Mg^{++} and slightly activated by Fe^{+++}. The *A. niger* enzyme was found to be more sensitive to pCMB, $CuSO_4$ and $HgCl_2$ as compared to the *Acetobacter suboxydans* enzyme (CUMMINS *et al.*, 1957).

K_m values for fructose and sorbitol were found to be 6.6×10^{-4} M and 9.8×10^{-5} M respectively. The double reciprocal plot for sorbitol is shown in Fig. 5. The K_m for sorbitol dehydrogenase purified from rat liver by Blackley (1951) was found to be 7×10^{-4} M for sorbitol. The K_m for fructose for mannitol dehydrogenase purified by Martinez *et al.* (1963) was found to be 0.07 M which is very high compared to sorbitol dehydrogenase from *A. niger*.

Acknowledgement. This work was supported by the Post-graduate Research Council of the M. S. University of Baroda.

References

Blackley, R. L.: The metabolism and antiketogenic effect of sorbitol. Sorbitol dehydrogenase. Biochem. J. **49**, 257 (1951).

Cummins, J. T., V. H. Cheldelin, and T. E. King: Sorbitol dehydrogenases in *Acetobacter suboxydans*. J. biol. Chem. **226**, 301 (1957).

Desai, B. M., V. V. Modi, and V. K. Shah: Induction of sorbitol dehydrogenase by sorbitol in *Aspergillus niger*. Arch. Mikrobiol. **56**, 300–304 (1967).

— — — Studies on polyol metabolism in *Aspergillus niger*. I. Nutritional requirements of a strain of *Aspergillus niger* cultivated on sorbitol as sole source of carbon. Arch. Mikrobiol. **67**, 6–11 (1969).

— — — Studies on polyol metabolism in *Aspergillus niger*. II. Comparative studies on the enzyme make-up of the adapted strain and parent strain. Arch. Mikrobiol. **67**, 12–15 (1969b).

Horwitz, S. B., and N. O. Kaplan: Hexitol dehydrogenase of *Bacillus subtilis*. J. biol. Chem. **239**, 830–838 (1964).

Klungsøyr, L.: Purification of Mannitol-1-phosphate dehydrogenase of *Escherichia coli*. Biochim. biophys. Acta (Amst.) **128**, 55–62 (1966).

Lewis, D. H., and D. C. Smith: Sugar alcohols (Polyols) in fungi and green plants. 1. Distribution, physiology and metabolism: New Phytologist **66**, 143–184 (1967).

Lowry, O. H., N. J. Rosenbrough, A. L. Farr, and R. J. Randall: Protein measurement with the folin phenol reagent. J. biol. Chem. **193**, 265 (1951).

Martinez, G., H. A. Barker, and B. L. Horecker: A specific mannitol dehydrogenase of *Lactobacillus brevis*. J. biol. Chem. **238**, 1598 (1963).

McCorkindale, J., and N. L. Edson: Polyol dehydrogenases. Biochem. J. **57**, 518 (1954).

Touster, O., and D. R. D. Shaw: Biochemistry of the acyclic polyols. Physiol. Rev. **42**, 181 (1962).

Wolff, J. B., and N. O. Kaplan: D. Mannitol-1-phosphate dehydrogenase from *Escherichia coli*. J. biol. Chem. **218**, 849–869 (1956).

B. M. Desai
V. V. Modi
V. K. Shah
Department of Microbiology
Faculty of Science
M.S. University of Baroda
Baroda, India